

Cyber Crime Investigation Support Documents

Document 1: Cyber Cell Cover Letter

To

The Cyber Crime Investigation Cell
[Concerned City / State]

Subject: Submission of AI-Assisted Blockchain Transaction Analysis for Suspected Crypto Fraud Investigation

Respected Sir / Madam,

We hereby submit this letter to formally present an intelligence-based blockchain transaction analysis conducted using our **Hybrid AI-Powered Blockchain Fraud Detection & Investigation System**.

The objective of this submission is to assist the Cyber Crime Cell in identifying, understanding, and investigating suspected fraudulent activities occurring through peer-to-peer (P2P) cryptocurrency transactions on public blockchains.

Our system does **not** perform enforcement actions. Instead, it provides: - Explainable forensic indicators - AI-assisted risk assessment - Structured evidence trails

These outputs are designed to support lawful investigation, FIR registration, and further action under applicable cyber and financial crime laws.

All findings are derived from publicly available blockchain data and internally logged transaction records, without breaching privacy or altering blockchain integrity.

We respectfully submit the attached annexure detailing: - Flagged transaction hashes - Reasons for suspicion - Risk categorization - AI-based probability assessment

We request the Cyber Crime Cell to review the annexure and take appropriate investigative steps as deemed necessary under law.

Submitted by:

Project Team – DeFiGuard / CryptoTraceChain
(Technical Demonstration Project)

Document 2: FIR Annexure – Blockchain Fraud Intelligence Report

Annexure Title

Annexure A: AI-Assisted Blockchain Transaction Intelligence Report

1. Case Context

This annexure provides a structured intelligence summary of cryptocurrency transactions flagged as potentially fraudulent based on: - Rule-based forensic analysis - Machine Learning (Random Forest) risk prediction

The purpose of this annexure is to support: - FIR documentation - Preliminary investigation - Evidence correlation

2. Data Source & Integrity

- Blockchain Type: Public Blockchain (P2P Crypto Transactions)
- Data Nature: Read-only, immutable transaction records
- Storage: Secure MongoDB evidence store
- Version Control: MongoDB document versioning (__v field)

No data modification or tampering is performed.

3. Fraud Detection Methodology

A. Rule-Based Forensic Indicators

Each transaction is evaluated against predefined forensic rules:

1. Multiple Small Transfers (Smurfing):

Repeated low-value transactions from a single sender within a short period.

2. Circular Transfers:

Funds moving between the same wallet pairs repeatedly, indicating layering or wash transactions.

3. High-Frequency Transactions:

Unusually rapid transaction execution from the same wallet.

4. Unusual Amount Deviation:

Transaction values deviating significantly from network averages.

Each triggered rule contributes to a cumulative **Risk Score**.

B. Machine Learning Risk Prediction

- Model Used: Random Forest Classifier
- Model Type: Supervised Machine Learning (AI-assisted)
- Input Features:
 - Transaction amount
 - Time of transaction
 - Day of week

The ML model outputs a **probability score** indicating how closely the transaction matches known fraudulent patterns.

4. Risk Classification Logic

Category	Description
Normal	No significant indicators detected
Watchlist	Medium risk – requires monitoring
Suspicious	High risk – investigation recommended

Final classification is determined by combining: - Rule-based Risk Score (explainable) - ML Risk Probability (pattern-based)

5. Explanation of Key Fields

- **Risk Score:** Numerical value (0–100) representing rule violations
 - **ML Risk Score:** AI-derived probability of fraud
 - **Reason:** Human-readable explanation of triggered indicators
 - **Status:** Final classification (Normal / Watchlist / Suspicious)
 - **__v:** Database versioning field ensuring evidence integrity
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6. Intended Use by Authorities

This annexure may be used for: - FIR registration - Wallet linkage analysis - Identification of laundering patterns - Requesting KYC details from exchanges - Initiating fund freeze requests through lawful channels

7. Declaration

This report is generated as a technical intelligence aid and does not claim guilt or liability. All enforcement decisions remain solely with the competent legal authorities.

End of Annexure